

Optimizing Inventory to Recover Lost Revenue

Analysis Based on Sales & Inventory Data

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Key Insights Summary

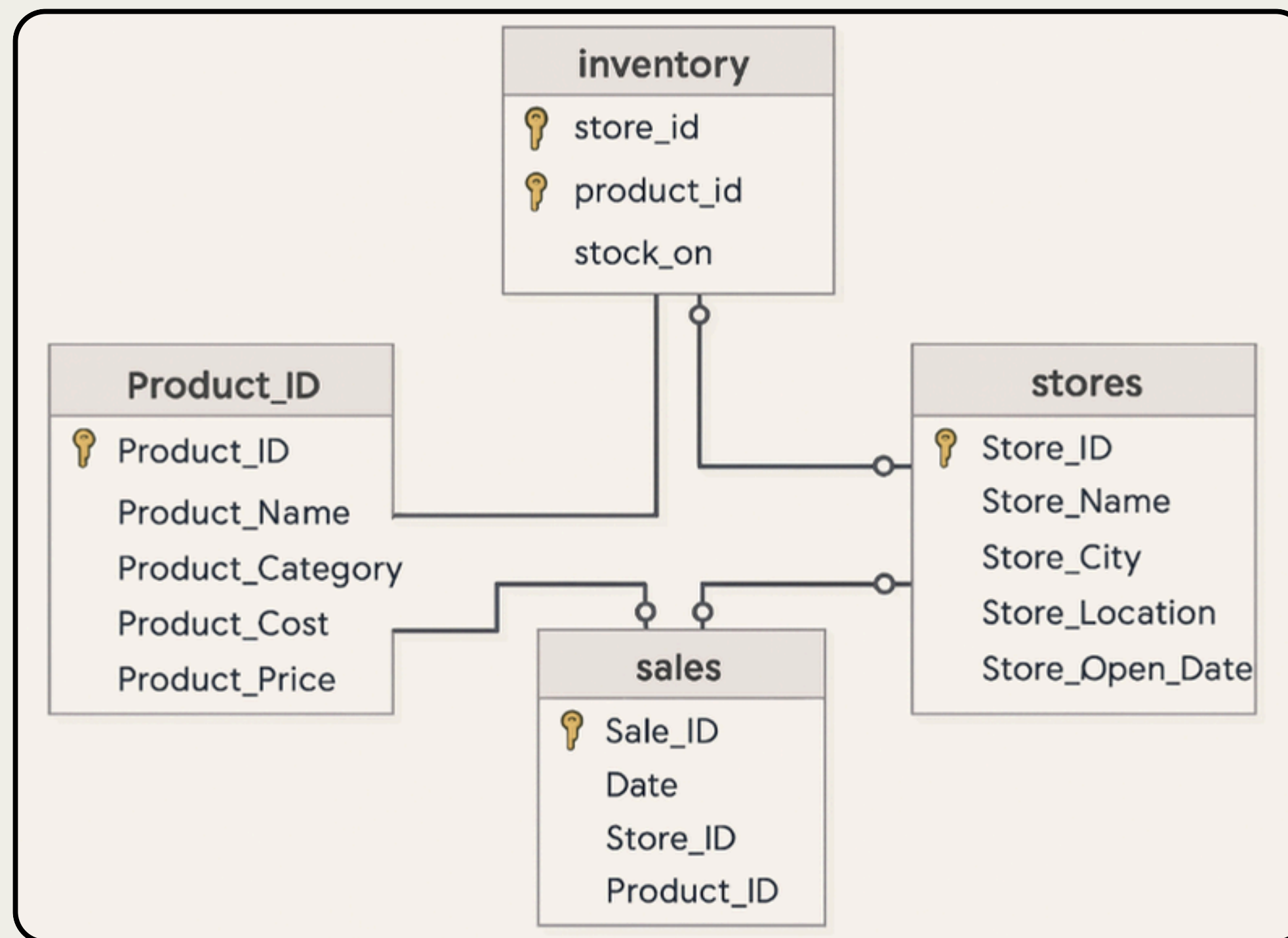


- 1. Inventory shortages cause lost revenue:**
≈190K dollars lost per year (≈4.7% of gross profit).
- 2. Five combinations account for over 43% of the losses.**
- 3. Excess stock is negligible:**
≈0.05% of yearly profit (mostly low-value items).
- 4. The opportunity is clear:**
Improve availability of high-demand, high-profit items.



DATA SET OVERVIEW

Data provides the foundation for demand analysis, profitability insights, and data-driven decisions.



Business Snapshot

50	Active Stores
35	Products
1.09M	Units Sold
829K	Transactions Completed
\$14.4M	Total Revenue
\$4.01M	Total Gross Profit
27.8%	Gross Profit Margin

Analysis Objective - Inventory Optimization

The core inventory problem:

1. High-Profit + High-Demand + Low Inventory



Missed Revenue

2. Low-Profit + Low-Demand + Excess Inventory

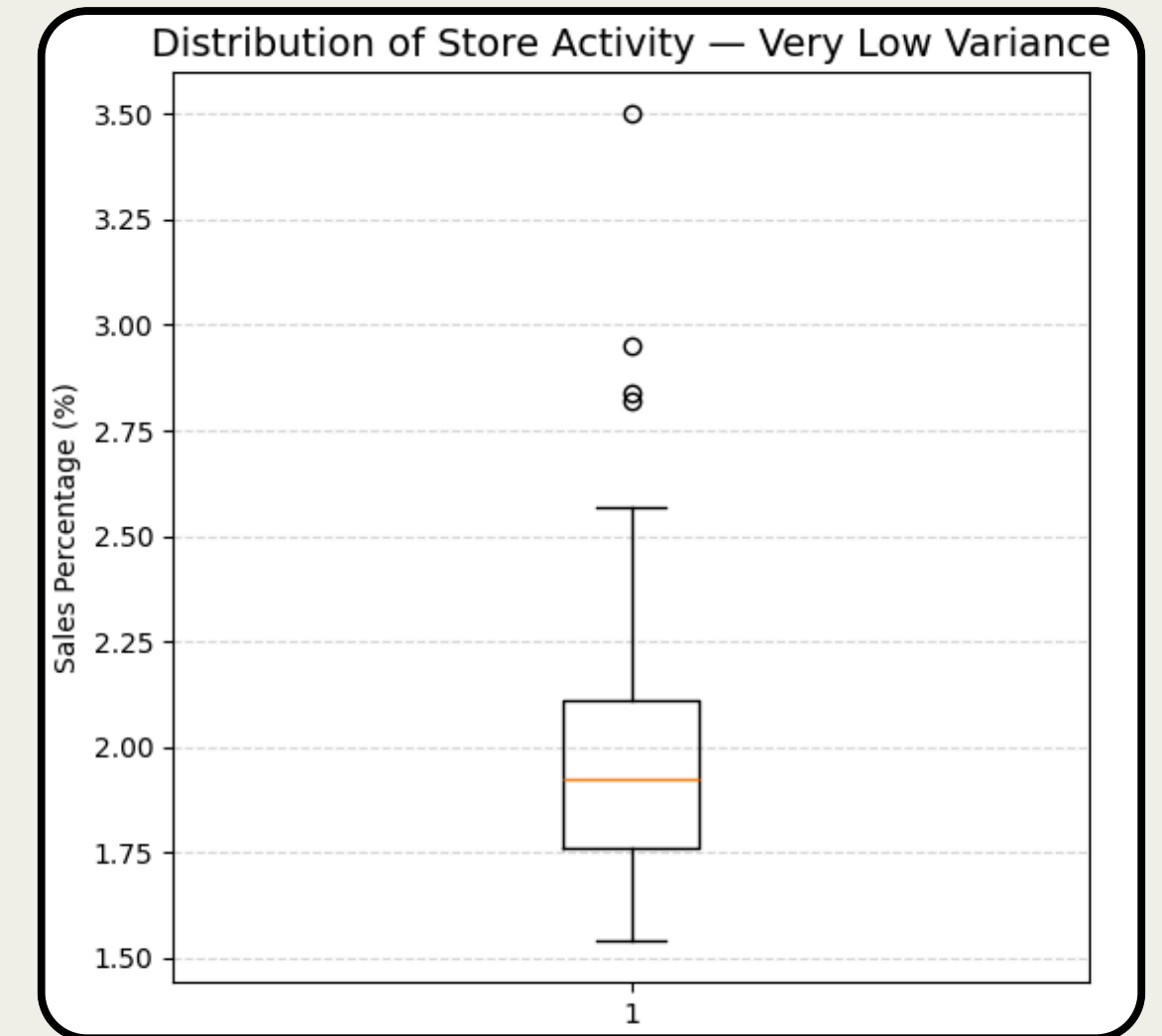
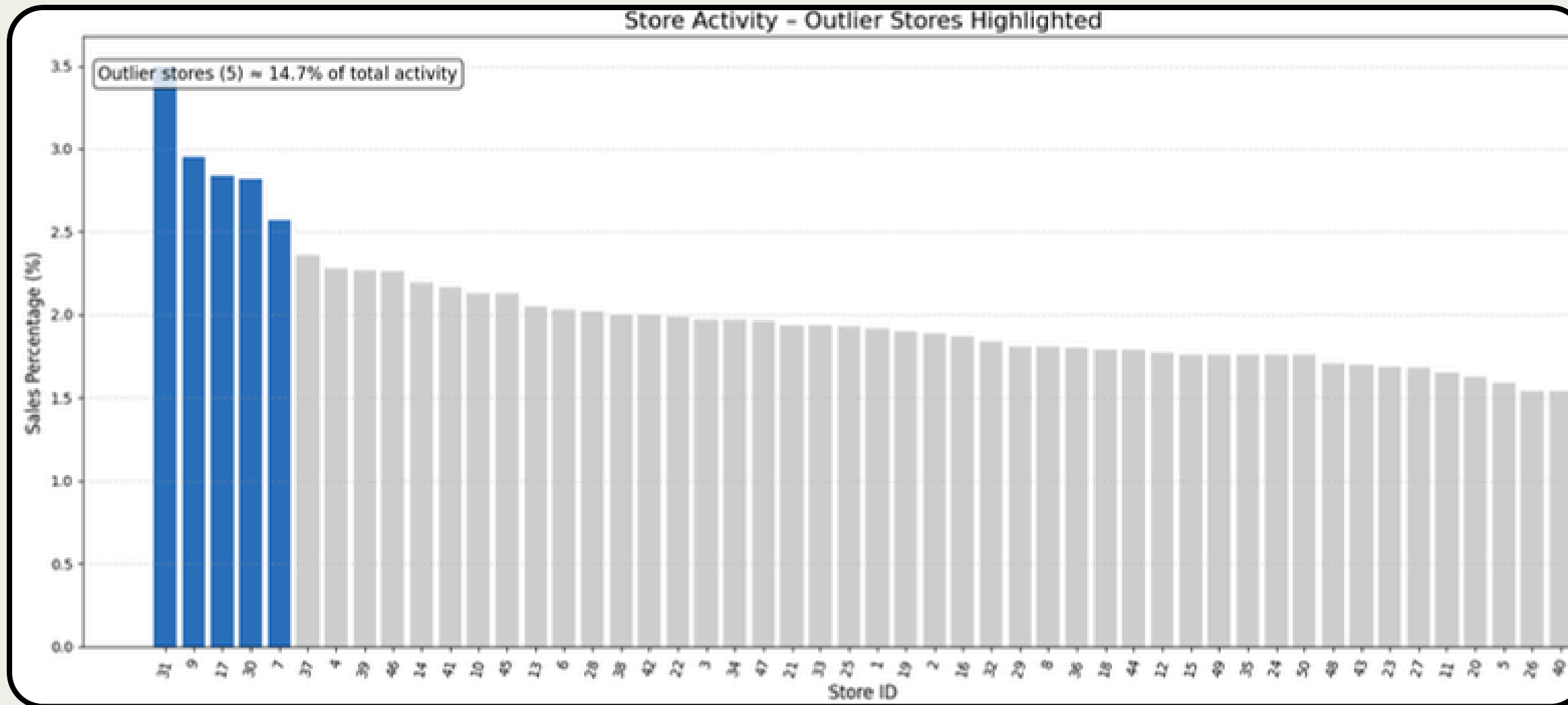


Tied-Up Capital

Focus on: top stores & high-profit products



Identifying the Most Active Stores



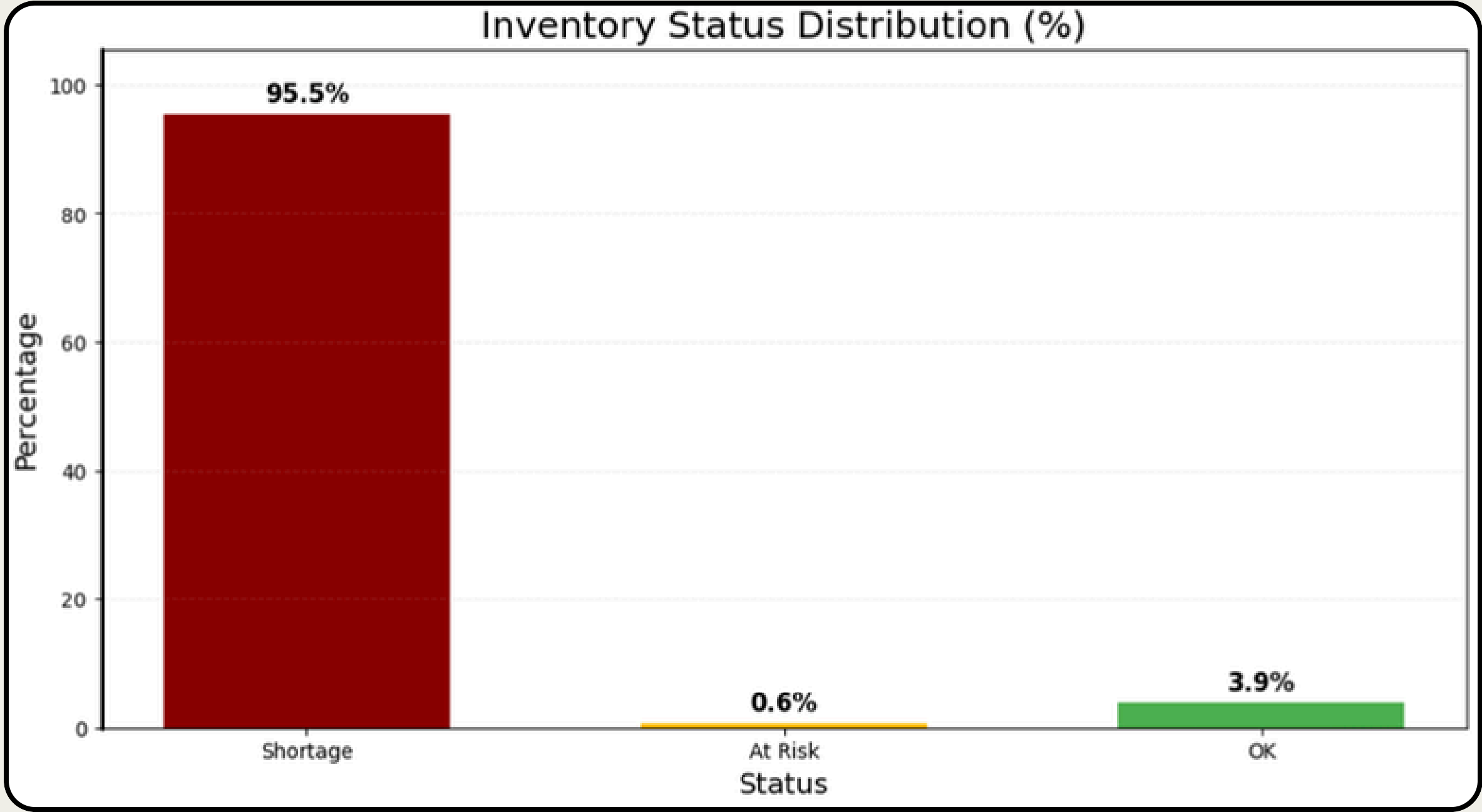
While activity is fairly uniform across stores, five locations stand out as clear top performers.

Most Profitable Products



Over 50% of total profit comes from just six products, the company's core profitability drivers.

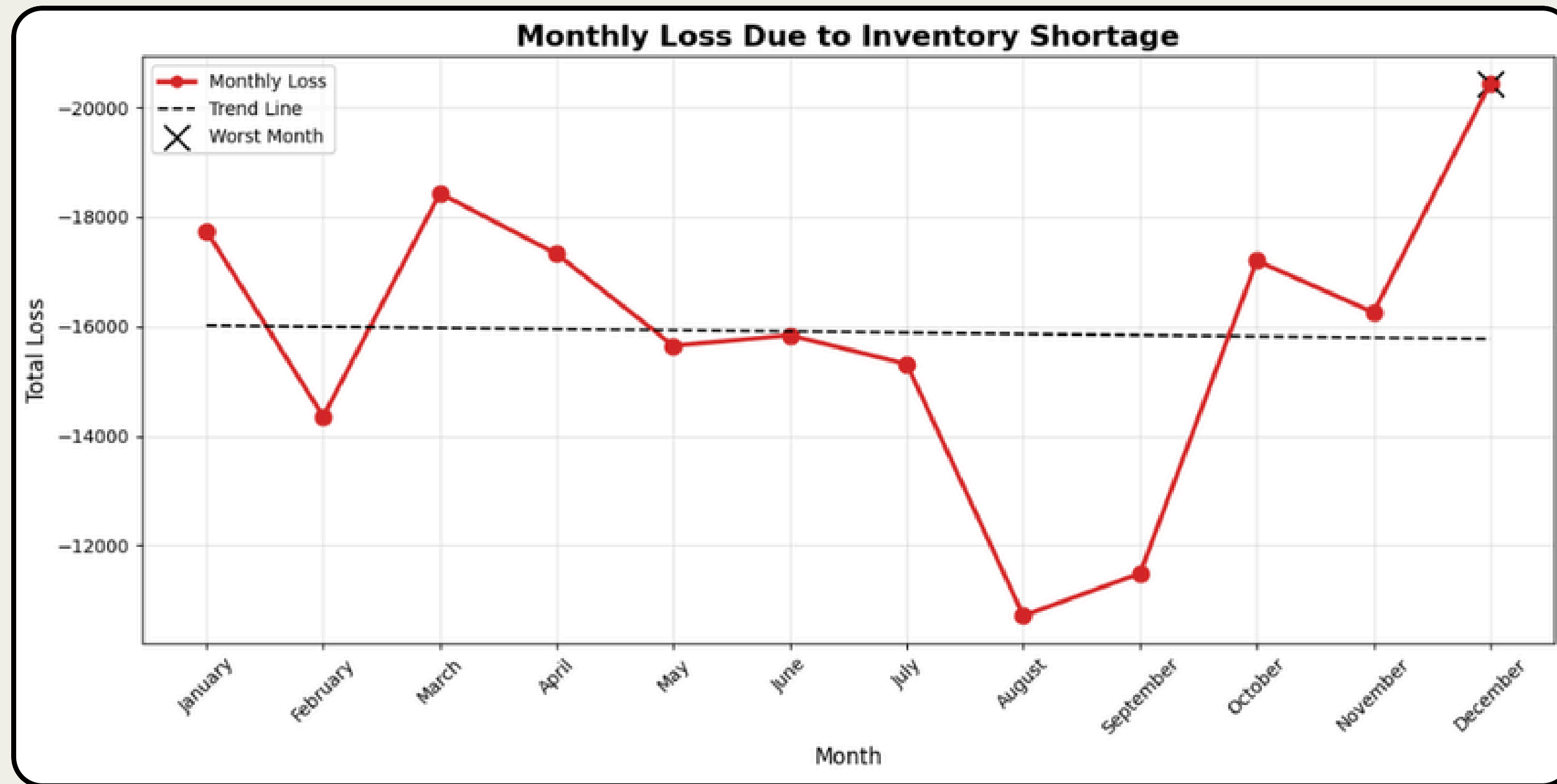
1. Potential Under-Stocking Risk



Almost all combinations fall below adjusted demand, indicating widespread understocking.

Based on current inventory vs. average monthly demand

How Much Revenue Could We Be Missing?

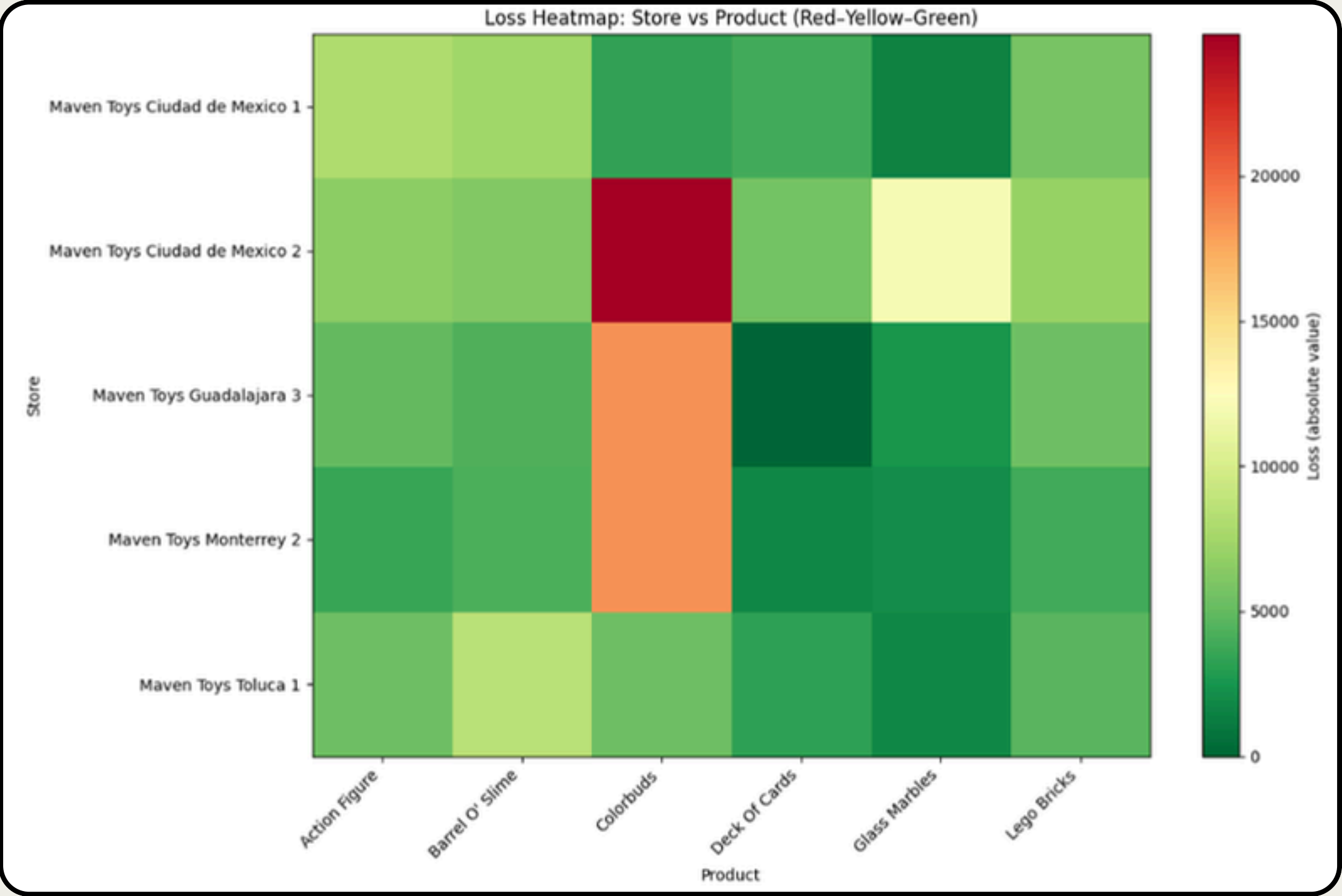


\$190K in potential
yearly revenue loss:
≈ 4.7% of gross profit.

Based on current inventory vs. average monthly demand

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Critical Loss Hotspots



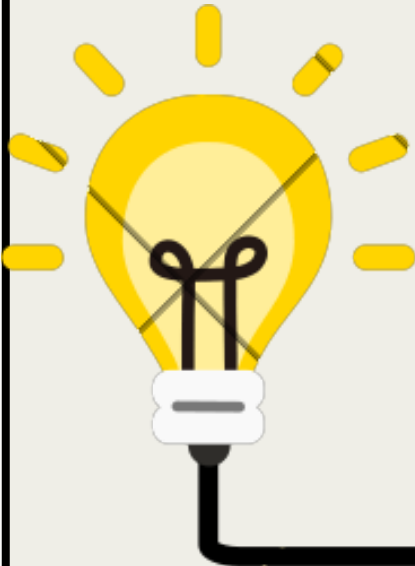
Revenue losses are highly concentrated:
a few key hotspots drive most of the impact

Based on current inventory vs. average monthly demand

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Critical Loss Hotspots

Product	Store	Loss (\$)
Colorbuds	Ciudad de Mexico	24,876
Colorbuds	Monterrey	18,820
Colorbuds	Guadalajara	18274
Glass Marbles	Ciudad de Mexico	11972
Barrel O' Slime	Toluca	8588
		81,990

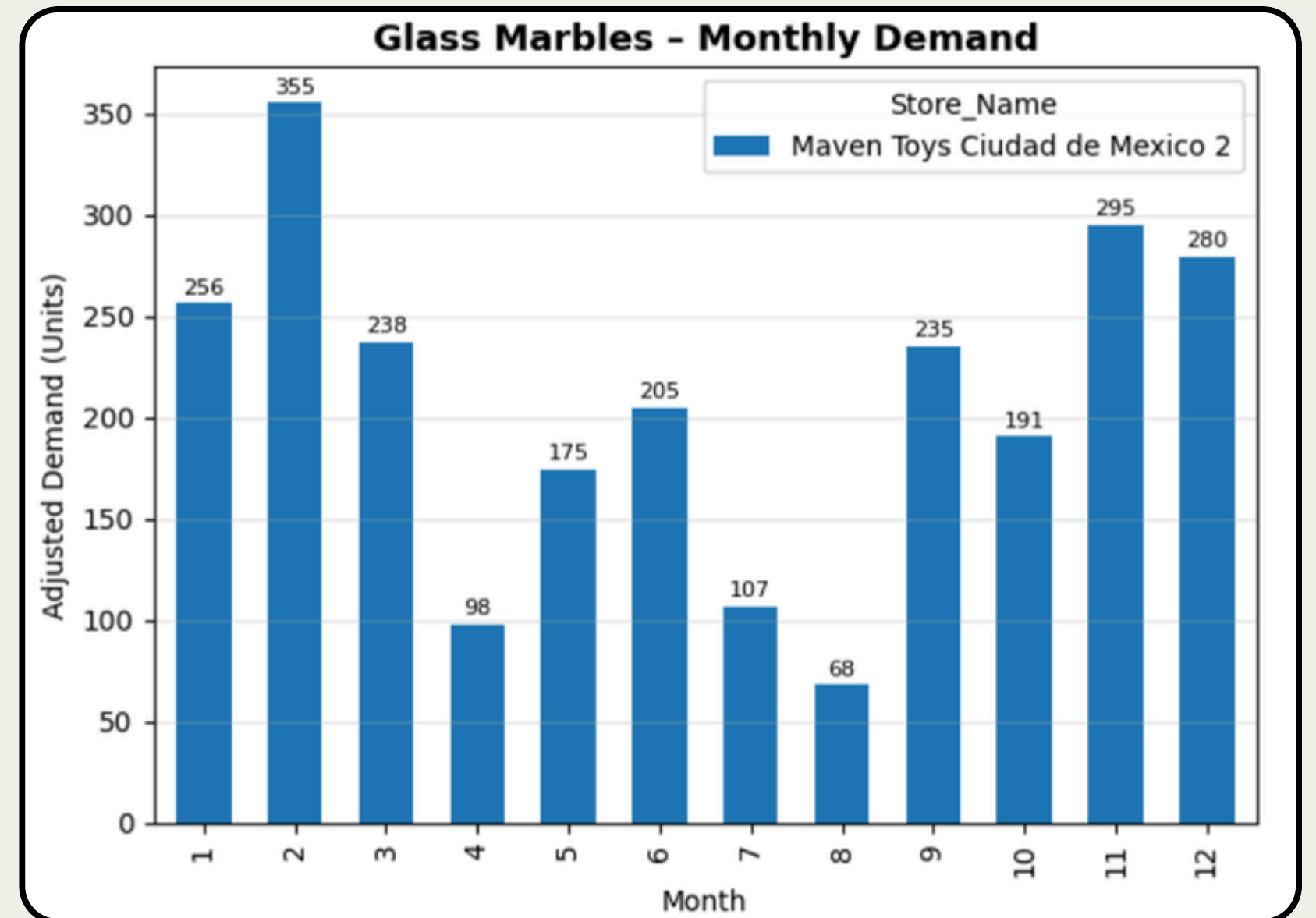
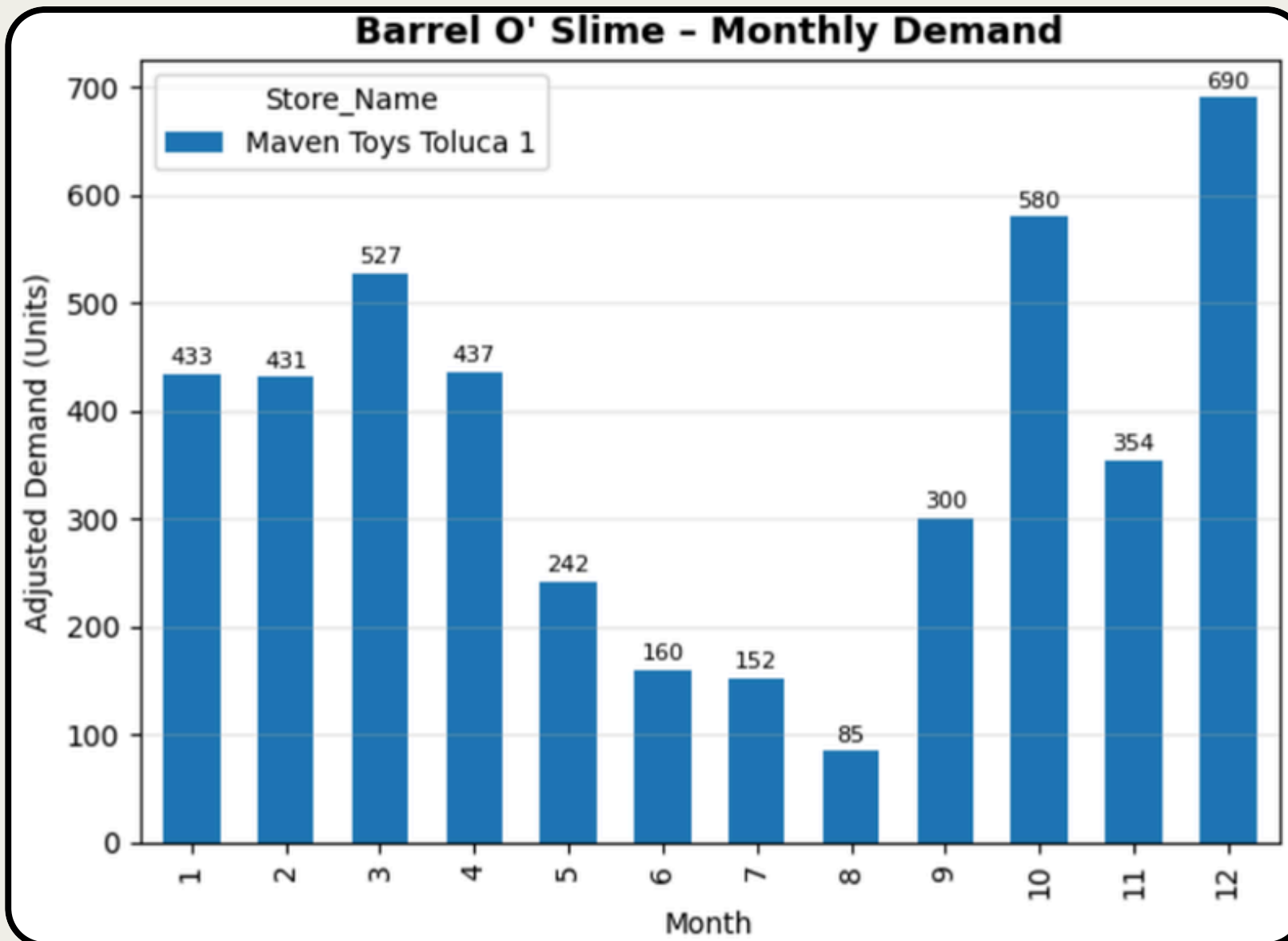


Fixing these key combinations could recover over 43% of potential revenue loss.

Based on current inventory vs. average monthly demand

Optimized Ordering

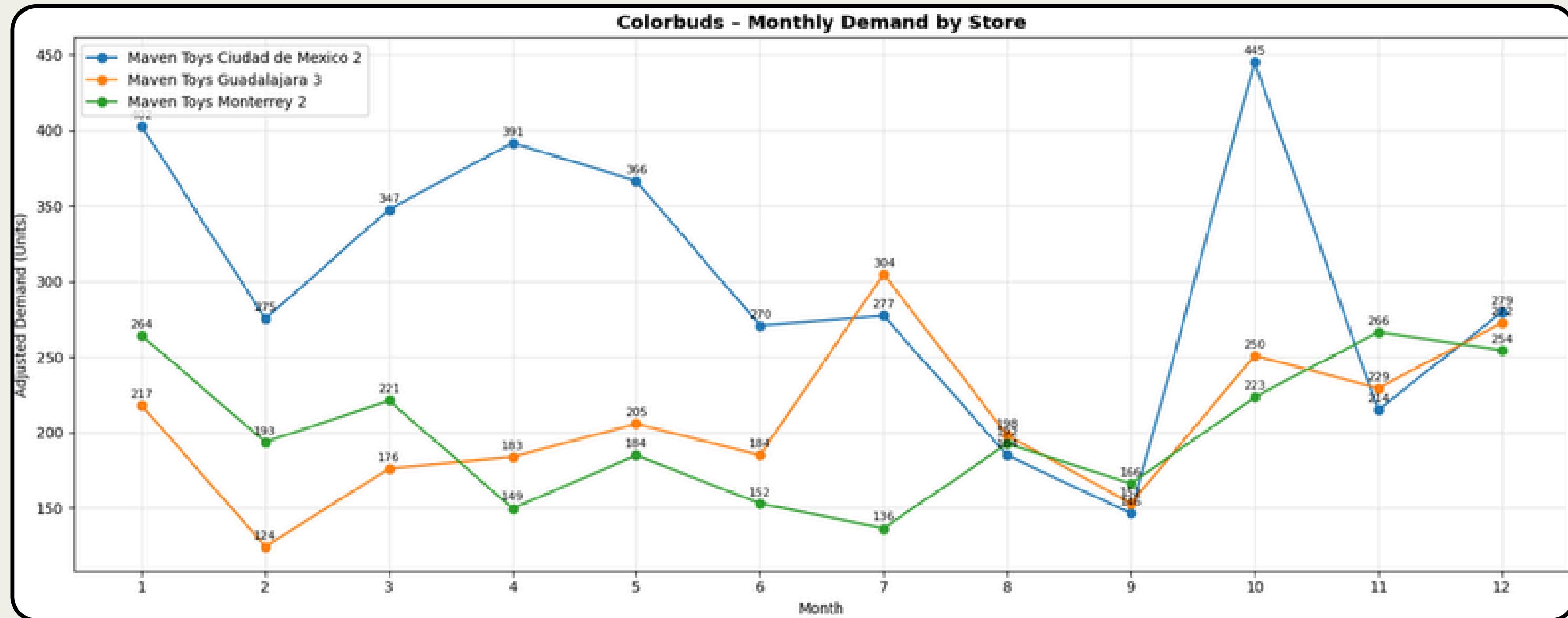
Ordering plan for the five critical combinations



This plan reduces lost sales and increases revenue.

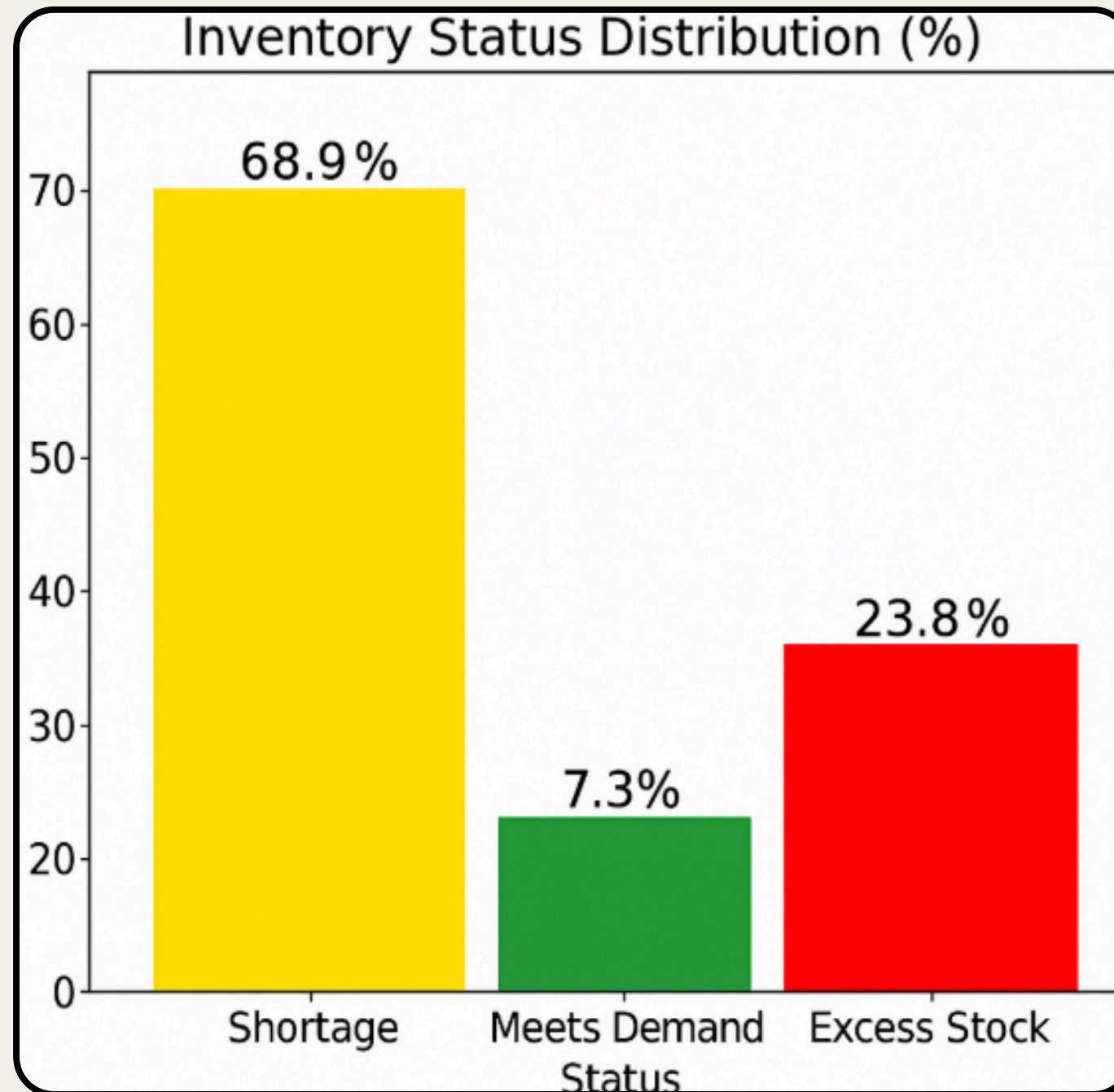
Optimized Ordering

Ordering plan for the five critical combinations



This plan reduces lost sales and increases revenue.

2. Potential Over-Stocking Risk

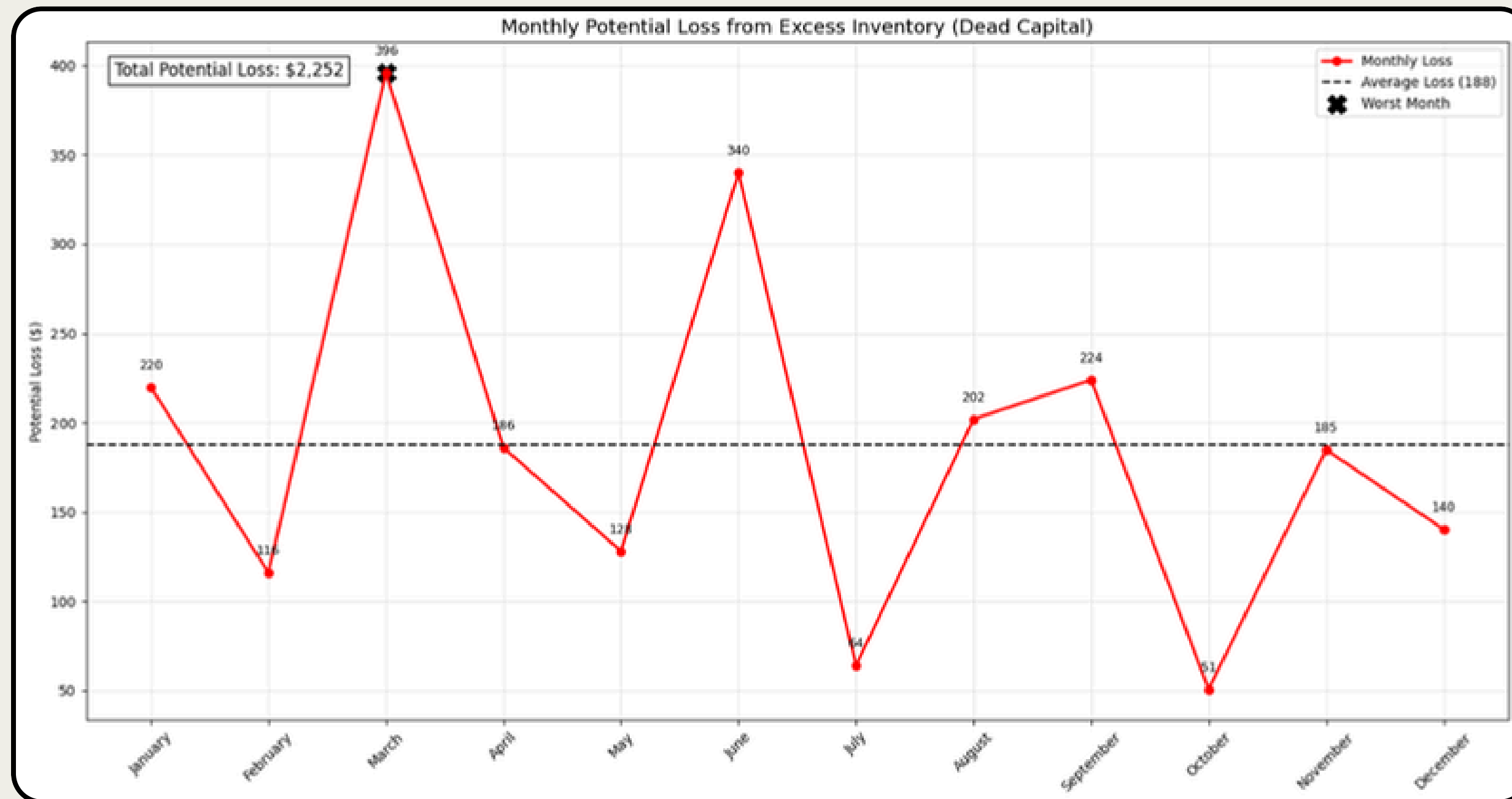


24% of inventory is potential excess stock driven by low demand and low profit.

Based on current inventory vs. average monthly demand



How much does excess stock cost?



The annual cost of excess stock is about 0.05% of yearly gross profit and is practically negligible.

Based on current inventory vs. average monthly demand

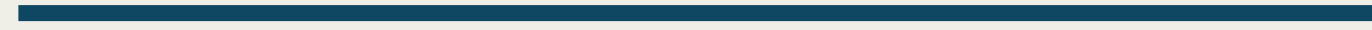


Executive Action Plan



- Identified 5 product-store combinations driving ~43% of potential revenue loss.
- Built an annual ordering plan to ensure monthly demand is met for these combinations.
- A yearly review is required to evaluate the plan's effectiveness and adjust based on performance.





Thank you

